

Tutorial on Peptide Generation Using MolAICal

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1. Introduction

This part summarizes three commonly encountered formats for representing peptide and biopolymer information: BILN (Biopolymer Line Notation)¹, HELM (Hierarchical Editing Language for Macromolecules)², and FASTA. Each format has different design goals and typical uses: one is lightweight and human-readable, one is a formal standard for macromolecules, and one is a minimal sequence container widely used in bioinformatics. Each format's defining features, typical use cases, strengths, and limitations are described, and given one clear example as follows:

FASTA is the legacy text-based format for representing nucleotide sequences or amino acid (protein) sequences. It is the universal standard for bioinformatics tools (e.g., BLAST) and simple sequence storage.

- **Limitations:** FASTA cannot inherently describe:
 - Cyclization (topology).
 - Branching or post-translational modifications (without non-standard hacks).
 - Complex chemical linkers or non-natural amino acids (unless 'X' is used ambiguously).

BILN and HELM can solve the limitations of FASTA. The BILN is the default format in MolAICal. Here, this tutorial will show the demo of peptide generations by MolAICal.

2. Materials

2.1. Software requirement

- 1) MolAICal: <https://molaical.github.io>
- 2) PyMol: <https://github.com/cgohlke/pymol-open-source-wheels> , <https://github.com/maabuu/pymol-wheels> , <https://pypi.org/project/pymol-open-source-whl/> , <https://github.com/cnpem/PyMOL4Win>

2.2. Example files

- 1) All the necessary tutorial files are downloaded from:
<https://github.com/MolAICal/tutorials/tree/master/027-peptideGen>

3. Procedure

3.1. Generate linear peptides

Open the tutorial material folder "027-peptideGen":

```
#> cd 027-peptideGen
```

First of all, users can see more help information for peptide generation in MolAICal by the below

commands:

```
#> molaical.exe -call run -c pepgen -i --help
```

1) Generate 10 random linear peptides with 8 residues with C- and N- caps without 3D structures.

```
#> molaical.exe -call run -c pepgen -i -l 8 -n 10 -o sequences.txt
```

2) If 3D structures without C- and N- caps are required besides the peptide sequences, run commands below:

```
#> molaical.exe -call run -c pepgen -i -l 8 -n 10 -3d -nc -o sequences.txt -dir results
```

Note:

- ◆ **-c:** It will generate the peptides if its value is "pepgen"
- ◆ **-i:** The command arguments of the program. It includes **all command-line arguments** of **external programs**, but the '-i' parameter must be the last one specified, while all other identifiers (e.g., '-p', etc.) of MolAICal must come **before '-i'**.
- ◆ **-l:** Number of amino acid residues in the peptide
- ◆ **-n:** Number of unique sequences to generate
- ◆ **-o:** Output file for generated sequences; The default file name is "peptide_sequences.txt".
- ◆ **-3d:** Generate 3D structures using pyPept
- ◆ **-dir:** Output directory for 3D structures; The default folder name is "output_3d".
- ◆ **-nc:** Do not add ac/am caps to sequences

For each peptide, MolAICal generates 5 types of peptides, which include **base**, **alpha-helix**, **beta-sheet**, **beta-bridge**, **turn-coil**, and **RDKit** formats. For example, "IVASETEE_turn-coil.pdb", the prefix "IVASETEE" is the sequence of this peptide. "turn-coil" is the structural type.

3.2. Generate cyclic peptides

Generate cyclic peptides; however, cyclic peptides are only supported in "base" and "turn-coil" formats, so the argument "-ss alpha-helix beta-sheet" will be ignored

```
#> molaical.exe -call run -c pepgen -i -l 8 -n 3 -cc -3d -c 4 -rc 3 -nc
```

Note:

- ◆ **-cc:** Generate cyclic peptides (BILN/HELM only)
- ◆ **-c:** Number of CPU cores for parallel processing (this "-c" is after "-i").
- ◆ **-nc:** Do not add ac/am caps to sequences
- ◆ **-rc:** Number of RDKit random conformers to generate (default: 1)

Open one cyclic peptide from the directory "output_3d" by PyMol (see Figure 1). Pymol does not show the cyclic cartoon peptide. Users can show line mode to show the cyclic peptide: "S" → "lines" (see Figure 1).



Figure 1

3.3. Modify peptides based on the fixed length

Randomly modify the peptide segment of length 16 using other amino acid residues, while keeping the amino acid residues at positions 1 and 3 unchanged

```
#> molaical.exe -call run -c pepgen -i -l 16 -n 10 -s CCCCCCCCCCCCCCCC -d none -fix 1 3 -f biln -3d -nc -c 10
```

Note:

- ◆ **-s:** Starting sequence (optional)
- ◆ **-d:** Direction for sequence extension from seed. If the value of "-d" is 'none', MolAICal will not extend the peptide.
- ◆ **-fix:** Positions (1-indexed) to keep fixed during generation
- ◆ **-f:** Output format for sequences (default: biln)

For more generated ways of peptides, please see the MolAICal manual.

References

1. Fox, T.; Bieler, M.; Haebel, P.; Ochoa, R.; Peters, S.; Weber, A., BILN: A Human-Readable Line Notation for Complex Peptides. *J. Chem. Inf. Model.* **2022**, *62* (17), 3942-3947.
2. Zhang, T. H.; Li, H. L.; Xi, H. L.; Stanton, R. V.; Rotstein, S. H., HELM: A Hierarchical Notation Language for Complex Biomolecule Structure Representation. *J. Chem. Inf. Model.* **2012**, *52* (10), 2796-2806.